Exp 4: Rutherford Scattering

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November 2024

1 Abstract

The objective of the Rutherford Scattering experiment is to measure the probability of alpha particles being scattered at different differential cross-sections and to compare the plot with the theoretical prediction. This experiment is based on the Rutherford scattering cross-section in α + Au elastic scattering. Experiment was conducted at the Edwards Accelerator Lab at Ohio University. Initially the energy calibration was done to verify that the peak positions matches with the kinematic relationships as expected. Afterwards the composition of the foil were calculated in terms of gold, silver and copper. The composition for gold was found to be 79.28%, 10.99% and 9.75%. Finally the experimental differential cross-section was obtained and was compared with the theoretical differential cross-section. It was seen that the two plots align with one another.
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2 Background Introduction

In 1911 Rutherford examined the absorption radioactivity of metal foil and found two components alpha radiation: absorbed by metal foil and beta radiation: able to pass through extremely thick metal foil before getting absorbed. Major observations of the Rutherford's alpha scattering experiment was observing a major portion of alpha particles passing through the sheet without any deflection: majority of the space in the atom is empty, some of the alpha particles were deflected by small angles and hence positive charge in an atom does not have uniform distribution and that the positive charge is concentrated at a small volume and lastly few alpha particles deflects back: positively charged particles is concentrated in a very small volume in comparison to the entire volume of the atom. This data led him to speculate the structure of the atom and obtain a newer nuclear atom model.
Scattering is an important concept as it gives details regarding the interactions between various particles and provides details regarding the internal structure of the atomic nuclei and its constituents. The basic concept is that the target is bombarded with a beam of particles with a different energies. The objective is to study the properties of the target when it's being bombarded by a beam of particles. A typical reaction,

$$a + b \rightarrow c + d$$

Here a and b are defined as the target and the beam of particles whereas c and d can be defined as the products of the reaction. When a beam approaches the target, a strong nuclear force between them would result in a deflection in the path of the target causing scattering. There are two kinds of scattering as elastic scattering and inelastic scattering. Elastic scattering is where the kinetic energy after collision would not change but the direction would. Inelastic scattering is where the beam looses kinetic energy during collision and transferring it to the target particle. In a scattering event there are various information that can be obtained about the dynamics of the interaction such as the coupling strength, shape of the interaction potential. The quantity that gives a proper understanding of these features is the cross-section (σ), probability of a reaction between two particles that are colliding.
The objective of this lab is to verify the structure of the atom by measuring how alpha particles are scattered

when being projected at a thin foil. This allows to determine the structure of the atomic nucleus by studying the angular distribution of the scattered particles. The differential calculation is calculated to obtain the probability an alpha particle will scatter into a specific angular range.
In this lab we first conducted the energy calibration to verify the peak positions with the kinematic relationship obtained. The the composition fo the foil was calculated in terms of gold, copper and silver. Finally the differential cross-section was plotted against the detector angles to verify with the theoretical differential cross-section.
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3 Theory

The scattering cross-section is the probability between the colliding particles. The geometric reaction cross-section is identified as the effective cross-sectional area of the target particles. Differential cross-section is a measurement of the probability of a particle that is scattering in a certain direction after a collision that can be expressed in terms of scattering rate per unit solid angle. In reality only a certain fraction of reactions can be measured. Considering that the area of the detector A_D and placed at a distance r, an angle θ with respective to the beam direction, then the solid angle: $\Delta\Omega = \frac{A_D}{r^2}$. Then the rate of reactions is proportional to the differential cross-section, $\frac{d\sigma(E,\theta)}{d\Omega}$,

$$N(E, \theta, \Delta\Omega) = L\frac{d\sigma(E, \theta)}{d\Omega}\Delta\Omega \tag{1}$$

The total cross-section σ is calculated by taking the integral over the total solid angle and overall scattering energies:

$$\sigma_{tot}(E) = \int_0^{E_{max}} \int_{4\pi} \frac{d^2\sigma(E, E', \theta)}{d\Omega dE'}\, d\Omega dE' \tag{2}$$

The Rutherford scattering cross-section can be derived classically or quantum mechanically using coulomb potential between two particles. Then the resulting differential cross-section is:

$$\frac{d\sigma(E, \theta)}{d\Omega_{cm}} = \frac{Z_0^2 Z_1^2 e^4}{16(4\pi\epsilon)^2 E_{cm}^2} \frac{1}{\sin^4(\theta_{cm}/2)} \tag{3}$$

E_{cm} and θ_{cm} the center of mass of kinetic energy and the scattering angle respectively.
Afterwards the coordinate system for the Rutherford formula needs to be converted to the Laboratory frame. This is done using the K factor.

$$k = \left(\frac{M_0 \cos(\theta) + \sqrt{M_i^2 - M_0^2 \sin^2(\theta)}}{M_0 + M_i} \right)^2 \tag{4}$$

M_0 is the atomic mass of helium and M_i is the atomic mass of the corresponding elements: Gold, Copper, Silver. θ is the laboratory scattering angle where 0 deg is considered to be the direction of the incident projectile. The differential cross-section in the laboratory is given by,

$$\frac{d\sigma}{d\Omega} = \frac{Z_0^2 Z_1^2 e^4}{4a(4\pi\epsilon)^2 E_0^2} \frac{(a + cos(\theta))^2}{\sin^4(\theta)} \tag{5}$$

E_0 is considered as the kinetic energy of the incident projectile. Energy of the scattered projectile is $E_0' = E_0 K$. K depends on the scattering angle and the target mass. After further simplifications the differential cross-section can be obtained as:

$$\frac{d\sigma}{d\Omega} = 1.296 \left(\frac{Z_0 Z_1}{E_0} \right)^2 \left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)} - 2\left(\frac{M_0}{M_1}\right)^2 \right) \tag{6}$$

This derivation of the formula yields nuclear physics units.
In this experiment the definition of E_0 is an approximation. A better approximation would be to take the average value of the energy considering the incident energy and the final energy (energy after passing through the gold foil).

$$E_{avg} = \frac{E_0 + E_f}{2} \tag{7}$$

$$E' = E_{avg} K \tag{8}$$

E_f is calculated taking the stopping power(Sp) into consideration. n is the target density, m is molar mass and A is avagadro number.

$$E_f = E_0 - \frac{Sp * n * m}{A} \tag{9}$$

The experimental differential cross-section can be stated as follows:

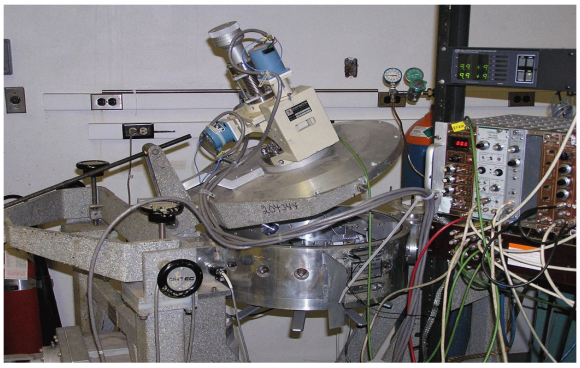
$$\frac{d\sigma}{d\Omega} = \frac{N_s}{N_0 \Delta\Omega n_1} \tag{10}$$

- N_s is the number of detected particles corrected by the RT/LT ratio. RT is Real time duration and LT is Live time duration.
- N_0 is the number of beam particles impinged on the scattering foil. This is obtained by the charge integrator.
- $\Delta\Omega$ is the solid angle.
- n_i is the number density per cm^2

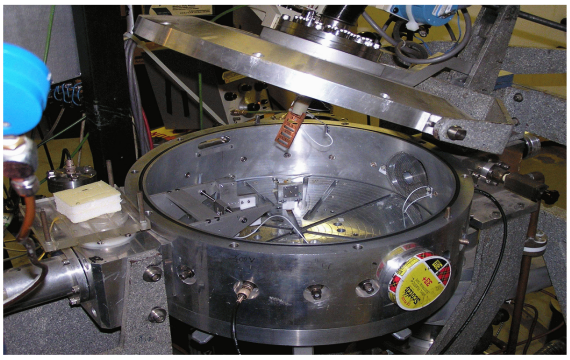
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4 Experimental Method

The experiment was conducted at the Edwards Accelerator Laboratory at Ohio University. The scattering chamber, is used to place a metal thin foil, allowing alpha particles to be fired at various angles without colliding with air molecules. The chamber is designed in a way that the detector angles could be placed precisely around the foil to count the number of scattered alpha particles at various directions.



(a) External view



(b) Internal view

Figure 1: Scattering chamber

Figure 2 represents the components of the scattering chamber. This is a vacuum chamber. A beam of

alpha particles are directed towards the target, gold foil where these alpha particles could pass. Gold was chosen as thin foils could be made, allowing alpha particles to interact with a minimum material as possible. The detector is positioned around the target to detect the scattering alphas. RBS is the fixed detector and PRD is the movable detector. The detector was places at different angles to measure the distribution of the scattered alphas. piDAQ is used to record the data from the detectors.

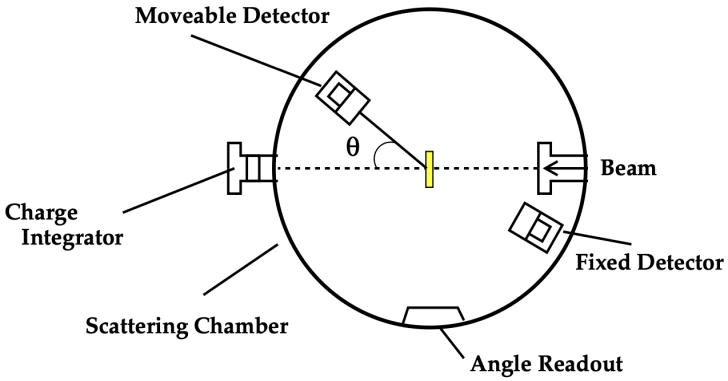


Figure 2: Components of the Scattering chamber

Initially the following data was collected that are needed for the calculations.

Table 1: Nominal experimental settings: constant energy

Type	Name	Description
RBS	Distance	6.700 ± 0.035 cm
	Diameter	4.01 mm
	Angle (Left)	222°335°
	Angle (Right)	25°155°
PRD	Distance	6.075 ± 0.035 cm
	Diameter	3.69 mm
	Angle (fixed)	168°

Table 2: Nominal experimental settings: constant angle

Type	Settings 1	Settings 2
θ_{PRD} (deg)	168	168
θ_{RBS} (deg)	50	155
θ_{tar} (deg)	25	0

In this experiment we recorded data at several scattering angles on either side of the incident beam direction. And then the counts were recorded for each silicon detectors.
This experiment was divided into two components, the angular variation with the differential cross-section and the energy variation with the differential cross-section. This report is solely conducted on the angular variation. The detector angle and the target angle was varied to a constant Energy of 2.5 MeV. BCI scale was denoted to be 10^{-8} and the PRD angle was set to at 168°. The current, dead times and the time duration was recorded for each tag file. Each run resulted in four output files as seen in Table 2.
Initially the analysis were done for the peaks of the RBS data file and then the gold peaks were identified. This was used to conduct the energy calibration and this was verified by calculating the residual energy and

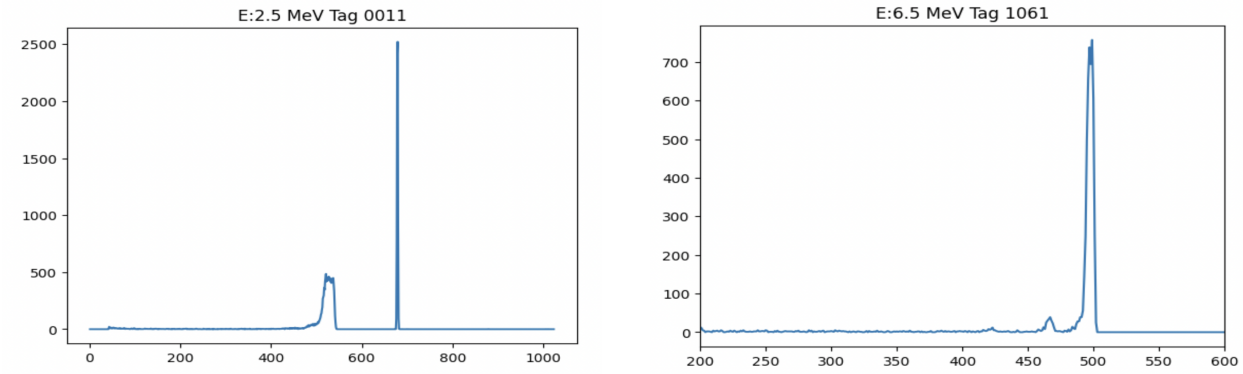
Table 3: Tag Description

Tag	output
Tag xxx0	PRD
Tag xxx1	RBS
Tag xxx2	LT/DT
Tag xxx3	BCI

by obtaining the bin numbers for the copper and silver peaks. It was seen that these bin numbers verify with the bin number obtained in the output files. Afterwards the composition of the foil was calculated in terms of gold, copper and silver. It was observed that the composition of gold as 79.28, 10.99, 9.72. Finally the differential cross-section was calculated using both experimental format and the theoretical format. This was calculated with a constant energy of 2.5 MeV and by varying the detector angle. It was observed that the two curves coincided one another proving that the experimental results align perfectly with the predictions made by Rutherford's theoretical model of atomic structure.
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5 Analysis and Results

The peak obtained from the RBS data files were analyzed and background subtraction were done to obtain a higher accuracy. It was observed that when the Energy is lower the separation of the peaks cannot be observed and only a superimposed peak can be obtained. But when the peaks of the higher energy data are analyzed the separation of gold, copper and silver peaks can be observed.



(a) Energy: 2.5 MeV

(b) Energy: 6.5 MeV

Figure 3: Peak Analysis

BS: Background Subtraction
Table 4 represents the data collected when the energy was kept at a constant. Table 5 shows the peak analysis done for different energies. Here we are only considering the gold peaks as its much more accurate to determine in respective to the other peaks. These energies and the bin numbers are used in energy calibration. Table 6 represents peak analysis conducted for the angle variation data where the energy was at constant of 2.5 MeV.

5.1 Energy Calibration:

Energy calibration is done according to Equation (8). Here the factor K is calculated for backward and forward angles for three different energy values. The average energy is taken in order to take into account

Table 4: Angular study nominal settings

Tag #	θ_{det} **(deg)**	θ_{tar} **(deg)**	**I (nA)**	**BCI**	**DTL**	**DTR**	**Time (sec)**
21	154.9	0.0	6	5000	1.0465	1.1628	86
31	60.0	30.0	6	5000	1.8391	1.3793	87
41	50.0	25.0	6	5000	5.4118	1.2941	85
51	40.0	20.0	6	5000	7.6136	1.25	88
61	315.0	337.5	6	5000	25.053	1.2629	95
71	305.0	332.5	6	5000	14.286	1.3187	91
81	295.0	327.5	6	5000	8.4043	1.276	94
91	270.0	315.0	6	5000	3.2673	1.2871	107
101	270.0	315.0	6	5000	3.8043	1.4130	92
111	245.0	0.0	6	5000	1.5217	1.1957	92
121	225.0	0.0	6	5000	1.1224	1.1224	98
131	325.0	342.5	6	5000	8.1818	0.56818	88
141	30.0	15.0	6	5000	10.28670	0.57143	105

Table 5: Peak Analysis: Energy variation

Tag #	**Energy (MeV)**	**Mean**	**RMS**	**Counts before BS**	**Counts after BS**	**Error**	**Precision**
11	3.5	279.74	2.15	64226.0	62740.0	0.01	0.39
21	3.5	262.64	3.16	5698.0	5328.0	0.04	1.32
31	4.5	339.56	6.37	3775.0	3671.0	0.10	1.63
41	4.5	362.38	4.30	3688.0	3062.0	0.07	1.65
51	6.5	527.30	6.19	4808.0	3527.0	0.09	1.44
61	6.5	496.92	3.22	4894.0	4586.0	0.05	1.43

Table 6: Peak Analysis: Angular variation

Tag #	**Mean**	**RMS**	**Counts before BS**	**Counts after BS**	**Error**	**Precision**
21	263.24	134.79	18087.0	12244.0	1.00	0.74
31	227.49	112.75	31517.0	21700.0	0.64	0.56
41	191.66	79.65	91912.0	65740.0	0.26	0.33
51	166.33	83.16	134572.0	70760.0	0.23	0.27
61	184.72	2.76	432733.0	420218.0	0.00	0.15
71	198.72	42.39	259947.0	234161.0	0.08	0.20
81	201.08	53.61	158712.0	140499.0	0.13	0.25
91	211.79	83.09	67149.0	57053.0	0.32	0.39
101	206.23	78.77	71853.0	62137.0	0.29	0.37
111	235.16	117.40	29186.0	21837.0	0.69	0.59
121	259.29	131.94	21942.0	15109.0	0.89	0.68
131	205.31	53.20	145931.0	131395.0	0.14	0.26
141	203.57	48.27	217642.0	197784.0	0.10	0.21

the energy loss of the alpha particles as they travel through the material. Therefore, using the beam energy E_0 would be an approximation since the energy of the alpha particles change as they go through the material. It would be more appropriate to take the average energy value as it accounts for this energy loss as well. Figure 4 represents the energy calibration conducted for the gold peaks for energies 3.5 MeV, 4.5 MeV and 6.5 MeV. Here the fitting function took the format of a linear relationship, y = mx + c.

$$y = \left(0.0118 \pm 5.7840 \times 10^{-5}\right) x + \left(0.1307 \pm 2.2743 \times 10^{-2}\right)$$

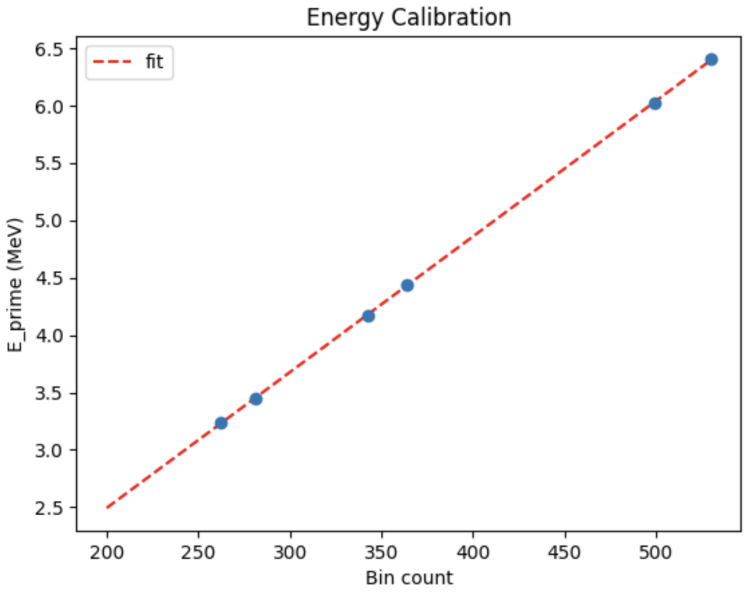


Figure 4: Energy Calibration

Then the residual energies were calculated to verify its accuracy.

Table 7: Residual Energy values

Angle (deg)	**Energy calculated (MeV)**	**Energy calibrated (MeV)**	**Residual (MeV)**
Forward angle (55)	3.45	3.45	0.0012
	4.44	4.43	-0.0038
	6.40	6.39	-0.0137
Backward angle (155)	3.24	3.23	-0.0130
	4.16	4.18	0.0185
	6.02	6.03	0.0106

Table 4 shows that the residual energy is extremely small verifying that the modeled function for the data values is accurate.

Table 8: Bin counts and Energy for Silver and Copper

Angle (deg)	**Energy: Cu**	**Bin counts: Cu**	**Energy: Ag**	**Bin counts: Ag**
Forward angle (55)	3.3459	272.1326	3.4084	277.4201
	4.3020	353.0475	4.3822	359.8459
	6.2140	514.8771	6.3300	524.6974
Backward angle (155)	2.7522	221.8841	3.0383	246.0935
	3.5386	288.4421	3.9064	319.5688
	5.1114	421.5580	5.6426	466.5192

Table 5 demonstrates the position of the peaks of silver and copper according to the obtained fit function. Comparing it with the data files, it was seen that the positions are extremely close. Therefore it can be concluded that the peak positions appear according to the prediction of the kinematic relationship.
Figure 5 shows the Energy values plotted against the residual energies.

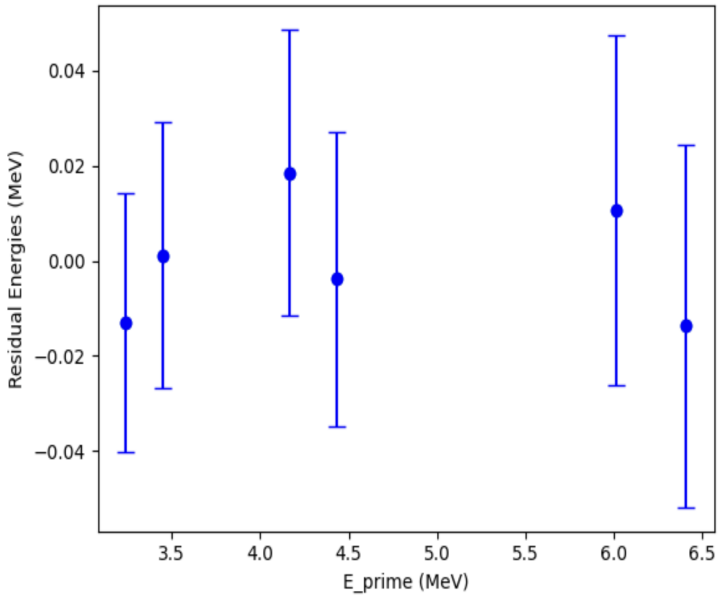


Figure 5: Energy Calibration

5.2 Composition of foil in terms of gold, silver and copper:

The composition was calculated for the back scattering angle: 155 deg at energy 6.5MeV. Theoretical differential cross-section is used in this context: Equation (6). Then using the number of counts we were able to derive a formula for the composition percentage for the three elements. The same process was conducted for copper and silver as well.

$$R_{AgAu} = \frac{N_A g}{N_A u} \frac{d\sigma_{Au}}{d\sigma_{Ag}}$$

$$R_{cuAu} = \frac{N_c u}{N_A u} \frac{d\sigma_{Au}}{d\sigma_{cu}}$$

$$Gold = (\frac{1}{1 + R_{AgAu} + R_{cuAu}})$$

Table 9: Composition of the foil

Element	Counts	Cross-section	Percentage	Error
Gold	4455.3	1091.55	79.2819	1.12075
Silver	218.74	385.68	10.9929	0.650352
Copper	73.1709	146.14	9.72527	1.29578

5.3 Differential cross-section vs angular distribution

Equation (6) and Equation (10) represents the theoretical differential cross-section and experimental differential cross-section respectively. Evaluating Figure 6 it can be observed that the theoretical and experimental

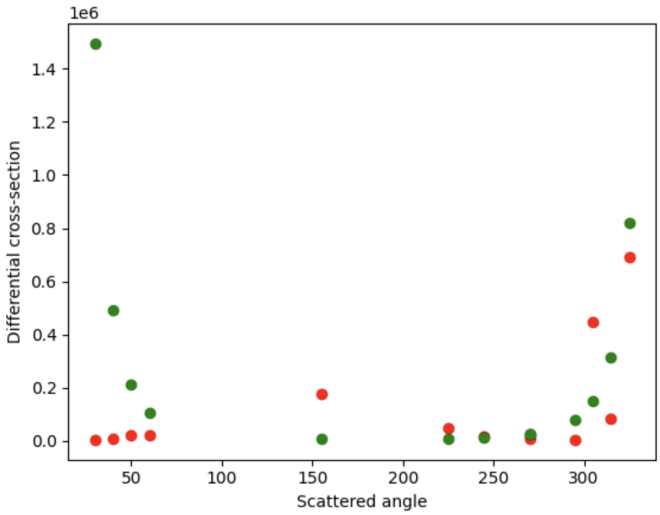


Figure 6: Angular Distribution

curves coincide with one another with a few exceptions. When the dead time ratio is higher then the observed count rates are much lower than the true value. This is due to the fact that the dead time ratio by definition means the time in which the detector does not record any event. When the count rate is high and there are more frequent events then there's a possibility that the detector might miss some events as its processing the previous events. This results in a fewer count loss. Therefore, when looking at the data the discrepancies between the analytical curve and experimental curve might be due to the higher dead-time rate in some data values.

Figure 7 demonstrates the ratio of the differential cross-section: experimental/theoretical against the scat-

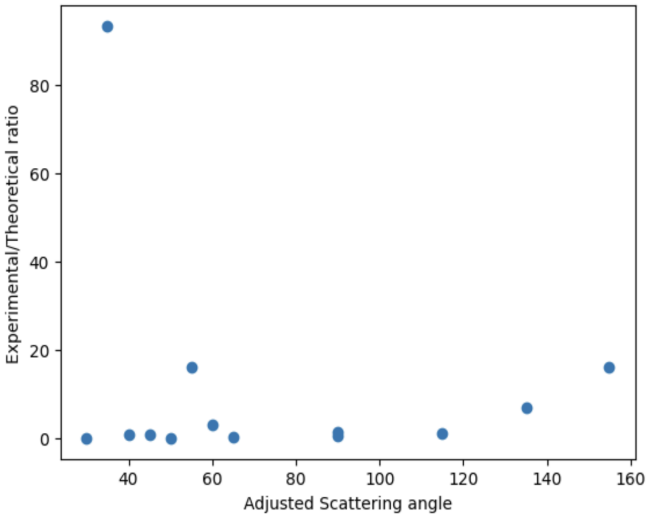


Figure 7: Angular Distribution

tered angle adjusted in the range of 0 to 180 deg. This gives an idea about the statistical uncertainty in the

measurements. It determines how well the experimental data matches the theoretical predictions.
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6 Conclusion

The Rutherford Scattering experiment was done to evaluate the experimental differential cross-section with the theoretical prediction. It can be said that the curves plotted align with one another with a few discrepancies due to the dead time correction. In Quantum mechanics the cross-section is calculated using the scattering amplitude and the formalism of quantum field theory. In the theoretical calculation the scattering amplitude determines the way the particles interact and in experimental measurements the number of particle interactions are taken into consideration.

7 References

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